

Supply Chain Analytics

SUPERSTORE DATASET — END-TO- END BUSINESS INTELLIGENCE PROJECT

Uncovering delivery performance gaps, quantity shortfalls, and profitability signals across 16,798 transactions. A complete BI solution built with Python, Power BI, and DAX.

GRADUATION PROJECT TEAM

WHAT BUSINESS CHALLENGE DOES THIS DATA ADDRESS?

⚠ CORE BUSINESS PROBLEM

The Global Superstore lacks unified visibility into its supply chain health, leading to undetected failures and missed performance targets.

01

NO UNIFIED VISIBILITY

No single view across sales, shipping, and delivery performance — data lives in silos.

02

LATE DELIVERIES

16.79% of orders delivered LATE — directly impacting customer satisfaction and retention.

03

QUANTITY SHORTFALLS

45.01% of orders have gaps between ordered and delivered quantity — undetected and untracked.

04

NO KPI FRAMEWORK

Only 44.6% of orders achieve full service level. Discount and margin patterns go unmonitored by segment and region.

🎯 Project Objective

Build an end-to-end BI solution to monitor supply chain performance in real time and identify root causes of delivery failures and quantity gaps.

Deliver actionable dashboards that drive data-driven decisions — shifting the organization from reactive firefighting to proactive performance management.

DATASET SCOPE, ORIGIN & STRUCTURE OVERVIEW

The Global Superstore dataset spans multi-year transactions across four global regions, covering sales, logistics, and financial performance at the line-item level.

16,798

TOTAL RECORDS

30

RAW COLUMNS

35

FINAL COLUMNS

4

GLOBAL REGIONS



GEOGRAPHY & DEPARTMENTS

4 Regions — AsiaPac, North America, EMEA, Latam. Departments: Technology · Furniture · Office Supplies.



CUSTOMER SEGMENTS & SHIPPING

Segments: Consumer · Corporate · Home Office · Small Business. Ship Modes: Regular Air · Express Air · Delivery Truck.



GRANULARITY

One row = One line-item order. Key fields: Sales, Profit, Discount, Unit Price, Shipping Cost, Order Quantity, Delivered Qty, Days To Ship.

PYTHON (PANDAS) — BEFORE VS. AFTER TRANSFORMATION PIPELINE



01 — DATE TYPE CONVERSION

Before: Order Date, Ship Date, Due Date stored as object/string.

After: Converted to datetime using `pd.to_datetime()` — enables date math and time intelligence.



02 — DUPLICATE & NULL CHECK

Before: Unchecked for duplicates. Postal Code & SubRegion null for non-NA regions.

After: 0 duplicate rows confirmed. Nulls documented as structural — not errors.



03 — DELIVERY STATUS ENGINEERING

Before: No field indicating if a shipment was late.

After: $\text{Days Difference} = \text{Ship Date} - \text{Due Date} \rightarrow \text{Status} = \text{'Late'} / \text{'On Time'}$



04 — QUANTITY GAP ENGINEERING

Before: No comparison between ordered vs. delivered quantity.

After: `deliver_difference` & `QTY_status` = 'Quantity Reached' / 'Not Reached'



05 — SERVICE LEVEL (ACHIEVED) FLAG

Before: No unified KPI combining delivery and quantity performance.

After: `Achieved` = 'achieved' only if **BOTH** on-time AND quantity reached — 44.6% baseline.

DAX FORMULAS POWERING THE POWER BI DASHBOARD KPIS

CALCULATIONS & MEASURES

TOTAL SALES

```
Total Sales = SUM(clean_superstore[Sales])
```

Aggregates total revenue across all transactions.

TOTAL PROFIT

```
Total Profit = SUM(clean_superstore[Profit])
```

Sum of net profit across all line items.

PROFIT MARGIN %

```
Profit Margin % = DIVIDE([Total Profit],[Total Sales])
```

Overall profitability ratio — currently **49.3%**.

ON-TIME DELIVERY %

```
On Time % =  
DIVIDE(COUNTROWS(FILTER(Table,[Status]="On  
Time")),COUNTROWS(Table))
```

Percentage of orders shipped by Due Date — **83.2%**.

QTY FULFILLMENT %

```
Qty Reached % =  
DIVIDE(COUNTROWS(FILTER(Table,[QTY_status]="Quantity  
Reached")),COUNTROWS(Table))
```

Orders where Delivered Qty = Order Quantity — **55.0%**.

SERVICE LEVEL %

```
Service Level =  
DIVIDE(COUNTROWS(FILTER(Table,[Achieved]="achieved")  
,COUNTROWS(Table))
```

Orders that are BOTH on-time AND full quantity — **44.6%**.

POWER BI FILE STRUCTURE — MODEL DESIGN & CHART SELECTIONS

Report Pages

01

EXECUTIVE SUMMARY

KPI cards + top-level overview across all regions and segments.

02

SALES ANALYSIS

Breakdown by region, segment, and category with time-series trends.

03

SUPPLY CHAIN PERFORMANCE

Delivery status, quantity fulfillment, and service level KPIs.

04

PROFITABILITY & DISCOUNT ANALYSIS

Margin matrix by region × segment, discount pattern exposure.

Chart Selections & Rationale

CLUSTERED BAR CHART

Sales by Region & Department —
easy multi-category comparison.

DONUT CHART

Order Status (On Time vs. Late) —
ideal for binary proportion display.

STACKED COLUMN

Sales over Time by Segment —
reveals trend and composition
together.

MAP VISUAL & TREEMAP

Geographic sales distribution +
hierarchical category contribution
at a glance.



Slicers: Region · Department · Segment · Ship Mode. Cross-filter
interactions enabled across all visuals.

DATA-DRIVEN FINDINGS EXTRACTED FROM 16,798 TRANSACTIONS

\$30.1M

TOTAL REVENUE

\$14.8M

TOTAL PROFIT

49.3%

PROFIT MARGIN

83.2%

ON-TIME RATE

44.6%

SERVICE LEVEL

Regional Performance

- AsiaPac leads with \$10.8M in sales (36% of total)
- North America follows at \$8.8M — highest profit margin
- Latam is lowest at \$4.9M — untapped growth potential

Department Breakdown

- Technology drives 41.6% of total sales (\$12.5M)
- Furniture is 2nd at \$10.0M — high shipping costs
- Office Supplies highest transaction volume, lowest ticket

Supply Chain Gaps

- 16.79% late delivery rate — 1 in 6 orders delayed
- 45% of orders have quantity shortfalls
- Only 44.6% achieve full service level (time + qty)

Profitability Signals

- Average discount: 5.7% — some segments over-discounted
- Regular Air = 51.9% of sales, most cost-effective mode
- Delivery Truck = highest cost, lowest profit per order

DATA-BACKED STRATEGIC ACTIONS FOR SUPPLY CHAIN IMPROVEMENT

1

CRITICAL

REDUCE LATE DELIVERIES FROM 16.8% → BELOW 5%

- Audit top delayed Ship Modes — Delivery Truck has highest late rate
- Set SLA alerts in Power BI for orders approaching Due Date
- Prioritize Critical & High priority orders for Express Air

2

HIGH

FIX QUANTITY FULFILLMENT (CURRENTLY 45% NOT REACHED)

- Investigate warehouse picking accuracy by region
- Flag customers and items with consistent shortfalls
- Set reorder thresholds for high-velocity SKUs

3

HIGH

GROW UNDERPERFORMING REGIONS (EMEA & LATAM = ONLY 34.7% OF SALES)

- Expand high-margin Technology category in EMEA and Latam
- Review shipping cost structure for cross-border Latam routes

4

MEDIUM

OPTIMIZE DISCOUNT STRATEGY (AVG 5.7%)

- Identify segments where discount >10% with negative profit impact
- Introduce discount caps by customer segment
- Redirect discount budget to high-margin Technology products

Q & A

Thank you for your attention

16,798

TRANSACTIONS ANALYZED

5

ENGINEERED KPIS

83.2%

ON-TIME DELIVERY RATE

44.6%

SERVICE LEVEL ACHIEVED

Data Source: Global Superstore · Tools: Python · Power BI · DAX · Graduation Project